# ECE250: Signals & Systems Quiz-2

**Date:** 03-September-2025 **Total Points:** 10 **Duration:** 15 Minutes

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**Note:**

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

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[CO1] **[Q1]**: We introduced a number of general properties of systems. In particular, a system may or may not be -

1. Time invariant
2. Linear
3. Causal
4. Stable

(a) Write the clear mathematical definition of all the above four properties. **[2 Points]**

(b) Determine which of these properties hold and which do not hold for each of the following continuous-time systems. Justify your answers. In each example, $y(t)$ denotes the system output and $x(t)$ is the system input.

(i) **[4 Points]**

$$y(t) = x(t-2) + x(2-t)$$

(ii) **[4 Points]**

$$y(t) = \int_{-\infty}^{2t} x(\tau) d\tau$$